$$
\begin{aligned}
& m_2^2 \to -C_{\phi 2^2} + \frac{1}{16\,\pi^2} \\
& \Bigg(-\frac{1}{4}\,C_{\phi 2^2}\,\left(g_1{}^2 + 3\,g_2{}^2\right) + \left(-\overline{C_{\phi 1\phi 2}}\,\tilde{\mu}\,\overline{y_e}^{pr}\,a_e{}^{pr} - \overline{a_e}^{pr}\,\left(C_{\phi 2^2}\,a_e{}^{pr} + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_e{}^{pr}\right)\right)\,LF_{2,1,0}\left[m_{\tilde{l}}{}^{p},\,m_{\tilde{e}}{}^{r}\right] + \\
& \quad \left(\overline{C_{\phi 1\phi 2}}\,\tilde{\mu}\,\overline{y_e}^{pr}\,a_e{}^{pr} + \overline{a_e}^{pr}\,\left(C_{\phi 2^2}\,a_e{}^{pr} + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_e{}^{pr}\right)\right)\,LF_{3,1,-1}\left[m_{\tilde{l}}{}^{p},\,m_{\tilde{e}}{}^{r}\right] + \\
& \quad \left(-\overline{C_{\phi 1\phi 2}}\,\tilde{\mu}\,\overline{y_e}^{pr}\,\left(a_e{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_e{}^{pr}\right) - \right. \\
& \qquad \left.\overline{a_e}^{pr}\,\left(a_e{}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,C_{\phi 1\phi 2} + C_{\phi 2^2}{}^2\right) + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_e{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2})\right)\right) \\
& \quad LF_{3,1,0}\left[m_{\tilde{l}}{}^{p},\,m_{\tilde{e}}{}^{r}\right] + 3\,\left(\overline{C_{\phi 1\phi 2}}\,\tilde{\mu}\,\overline{y_e}^{pr}\,\left(a_e{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_e{}^{pr}\right) + \right. \\
& \qquad \left.\overline{a_e}^{pr}\,\left(a_e{}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,C_{\phi 1\phi 2} + C_{\phi 2^2}{}^2\right) + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_e{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2})\right)\right) \\
& \quad LF_{4,1,-1}\left[m_{\tilde{l}}{}^{p},\,m_{\tilde{e}}{}^{r}\right] - 2\,\left(\overline{C_{\phi 1\phi 2}}\,\tilde{\mu}\,\overline{y_e}^{pr}\,\left(a_e{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_e{}^{pr}\right) + \right. \\
& \qquad \left.\overline{a_e}^{pr}\,\left(a_e{}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,C_{\phi 1\phi 2} + C_{\phi 2^2}{}^2\right) + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_e{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2})\right)\right)\,LF_{5,1,-2}\left[m_{\tilde{l}}{}^{p},\,m_{\tilde{e}}{}^{r}\right] - \\
& \quad 3\,\left(\overline{C_{\phi 1\phi 2}}\,\tilde{\mu}\,\overline{y_d}^{pr}\,a_d{}^{pr} + \overline{a_d}^{pr}\,\left(C_{\phi 2^2}\,a_d{}^{pr} + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_d{}^{pr}\right)\right)\,LF_{2,1,0}\left[m_{\tilde{q}}{}^{p},\,m_{\tilde{d}}{}^{r}\right] + \\
& \quad 3\,\left(\overline{C_{\phi 1\phi 2}}\,\tilde{\mu}\,\overline{y_d}^{pr}\,a_d{}^{pr} + \overline{a_d}^{pr}\,\left(C_{\phi 2^2}\,a_d{}^{pr} + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_d{}^{pr}\right)\right)\,LF_{3,1,-1}\left[m_{\tilde{q}}{}^{p},\,m_{\tilde{d}}{}^{r}\right] - \\
& \quad 3\,\left(\overline{C_{\phi 1\phi 2}}\,\tilde{\mu}\,\overline{y_d}^{pr}\,\left(a_d{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_d{}^{pr}\right) + \right. \\
& \qquad \left.\overline{a_d}^{pr}\,\left(a_d{}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,C_{\phi 1\phi 2} + C_{\phi 2^2}{}^2\right) + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_d{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2})\right)\right) \\
& \quad LF_{3,1,0}\left[m_{\tilde{q}}{}^{p},\,m_{\tilde{d}}{}^{r}\right] + 9\,\left(\overline{C_{\phi 1\phi 2}}\,\tilde{\mu}\,\overline{y_d}^{pr}\,\left(a_d{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_d{}^{pr}\right) + \right. \\
& \qquad \left.\overline{a_d}^{pr}\,\left(a_d{}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,C_{\phi 1\phi 2} + C_{\phi 2^2}{}^2\right) + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_d{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2})\right)\right) \\
& \quad LF_{4,1,-1}\left[m_{\tilde{q}}{}^{p},\,m_{\tilde{d}}{}^{r}\right] - 6\,\left(\overline{C_{\phi 1\phi 2}}\,\tilde{\mu}\,\overline{y_d}^{pr}\,\left(a_d{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_d{}^{pr}\right) + \right. \\
& \qquad \left.\overline{a_d}^{pr}\,\left(a_d{}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,C_{\phi 1\phi 2} + C_{\phi 2^2}{}^2\right) + C_{\phi 1\phi 2}\,\tilde{\mu}\,y_d{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2})\right)\right)\,LF_{5,1,-2}\left[m_{\tilde{q}}{}^{p},\,m_{\tilde{d}}{}^{r}\right] - \\
& \quad 3\,\tilde{\mu}\,\left(C_{\phi 1\phi 2}\,y_u{}^{pr}\,\overline{a_u}^{pr} + \overline{y_u}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,a_u{}^{pr} + C_{\phi 2^2}\,\tilde{\mu}\,y_u{}^{pr}\right)\right)\,LF_{2,1,0}\left[m_{\tilde{u}}{}^{r},\,m_{\tilde{q}}{}^{p}\right] + \\
& \quad 3\,\tilde{\mu}\,\left(C_{\phi 1\phi 2}\,y_u{}^{pr}\,\overline{a_u}^{pr} + \overline{y_u}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,a_u{}^{pr} + C_{\phi 2^2}\,\tilde{\mu}\,y_u{}^{pr}\right)\right)\,LF_{3,1,-1}\left[m_{\tilde{u}}{}^{r},\,m_{\tilde{q}}{}^{p}\right] - \\
& \quad 3\,\left(C_{\phi 1\phi 2}\,\overline{a_u}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,a_u{}^{pr} + \tilde{\mu}\,y_u{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2})\right) + \right. \\
& \qquad \left.\tilde{\mu}\,\overline{y_u}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,a_u{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2}) + \tilde{\mu}\,y_u{}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,C_{\phi 1\phi 2} + C_{\phi 2^2}{}^2\right)\right)\right) \\
& \quad LF_{3,1,0}\left[m_{\tilde{u}}{}^{r},\,m_{\tilde{q}}{}^{p}\right] + 9\,\left(C_{\phi 1\phi 2}\,\overline{a_u}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,a_u{}^{pr} + \tilde{\mu}\,y_u{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2})\right) + \right. \\
& \qquad \left.\tilde{\mu}\,\overline{y_u}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,a_u{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2}) + \tilde{\mu}\,y_u{}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,C_{\phi 1\phi 2} + C_{\phi 2^2}{}^2\right)\right)\right) \\
& \quad LF_{4,1,-1}\left[m_{\tilde{u}}{}^{r},\,m_{\tilde{q}}{}^{p}\right] - 6\,\left(C_{\phi 1\phi 2}\,\overline{a_u}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,a_u{}^{pr} + \tilde{\mu}\,y_u{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2})\right) + \right. \\
& \qquad \left.\tilde{\mu}\,\overline{y_u}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,a_u{}^{pr}\,(C_{\phi 1^2} + C_{\phi 2^2}) + \tilde{\mu}\,y_u{}^{pr}\,\left(\overline{C_{\phi 1\phi 2}}\,C_{\phi 1\phi 2} + C_{\phi 2^2}{}^2\right)\right)\right)\,LF_{5,1,-2}\left[m_{\tilde{u}}{}^{r},\,m_{\tilde{q}}{}^{p}\right] + \\
& \quad \frac{3}{2}\,C_{\phi 2^2}\,g_1{}^2\,LF_{2,1,-1}[\tilde{\mu},\,m_1] - m_1\,\tilde{\mu}\,g_1{}^2\,\left(\overline{C_{\phi 1\phi 2}} + C_{\phi 1\phi 2}\right)\,LF_{2,1,0}[\tilde{\mu},\,m_1] - C_{\phi 2^2}\,g_1{}^2\,LF_{3,1,-2}[\tilde{\mu},\,m_1] + \\
& \quad g_1{}^2\,\left(4\,\overline{C_{\phi 1\phi 2}}\,C_{\phi 1\phi 2} + 2\,C_{\phi 2^2}{}^2 + m_1\,\tilde{\mu}\,\left(\overline{C_{\phi 1\phi 2}} + C_{\phi 1\phi 2}\right)\right)\,LF_{3,1,-1}[\tilde{\mu},\,m_1] - \\
& \quad m_1\,\tilde{\mu}\,g_1{}^2\,\left(\overline{C_{\phi 1\phi 2}} + C_{\phi 1\phi 2}\right)\,(C_{\phi 1^2} + C_{\phi 2^2})\,LF_{3,1,0}[\tilde{\mu},\,m_1] - \\
& \quad 4\,g_1{}^2\,\left(2\,\overline{C_{\phi 1\phi 2}}\,C_{\phi 1\phi 2} + C_{\phi 2^2}{}^2\right)\,LF_{4,1,-2}[\tilde{\mu},\,m_1] + \\
& \quad 3\,m_1\,\tilde{\mu}\,g_1{}^2\,\left(\overline{C_{\phi 1\phi 2}} + C_{\phi 1\phi 2}\right)\,(C_{\phi 1^2} + C_{\phi 2^2})\,LF_{4,1,-1}[\tilde{\mu},\,m_1] + \\
& \quad 2\,g_1{}^2\,\left(2\,\overline{C_{\phi 1\phi 2}}\,C_{\phi 1\phi 2} + C_{\phi 2^2}{}^2\right)\,LF_{5,1,-3}[\tilde{\mu},\,m_1] - \\
& \quad 2\,m_1\,\tilde{\mu}\,g_1{}^2\,\left(\overline{C_{\phi 1\phi 2}} + C_{\phi 1\phi 2}\right)\,(C_{\phi 1^2} + C_{\phi 2^2})\,LF_{5,1,-2}[\tilde{\mu},\,m_1] + \frac{9}{2}\,C_{\phi 2^2}\,g_2{}^2\,LF_{2,1,-1}[\tilde{\mu},\,m_2] - \\
& \quad 3\,m_2\,\tilde{\mu}\,g_2{}^2\,\left(\overline{C_{\phi 1\phi 2}} + C_{\phi 1\phi 2}\right)\,LF_{2,1,0}[\tilde{\mu},\,m_2] - 3\,C_{\phi 2^2}\,g_2{}^2\,LF_{3,1,-2}[\tilde{\mu},\,m_2] + \\
& \quad 3\,g_2{}^2\,\left(4\,\overline{C_{\phi 1\phi 2}}\,C_{\phi 1\phi 2} + 2\,C_{\phi 2^2}{}^2 + m_2\,\tilde{\mu}\,\left(\overline{C_{\phi 1\phi 2}} + C_{\phi 1\phi 2}\right)\right)\,LF_{3,1,-1}[\tilde{\mu},\,m_2] - \\
& \quad 3\,m_2\,\tilde{\mu}\,g_2{}^2\,\left(\overline{C_{\phi 1\phi 2}} + C_{\phi 1\phi 2}\right)\,(C_{\phi 1^2} + C_{\phi 2^2})\,LF_{3,1,0}[\tilde{\mu},\,m_2] - \\
& \quad 12\,g_2{}^2\,\left(2\,\overline{C_{\phi 1\phi 2}}\,C_{\phi 1\phi 2} + C_{\phi 2^2}{}^2\right)\,LF_{4,1,-2}[\tilde{\mu},\,m_2] + \\
& \quad 9\,m_2\,\tilde{\mu}\,g_2{}^2\,\left(\overline{C_{\phi 1\phi 2}} + C_{\phi 1\phi 2}\right)\,(C_{\phi 1^2} + C_{\phi 2^2})\,LF_{4,1,-1}[\tilde{\mu},\,m_2] + \\
& \quad 6\,g_2{}^2\,\left(2\,\overline{C_{\phi 1\phi 2}}\,C_{\phi 1\phi 2} + C_{\phi 2^2}{}^2\right)\,LF_{5,1,-3}[\tilde{\mu},\,m_2] - \\
& \quad 6\,m_2\,\tilde{\mu}\,g_2{}^2\,\left(\overline{C_{\phi 1\phi 2}} + C_{\phi 1\phi 2}\right)\,(C_{\phi 1^2} + C_{\phi 2^2})\,LF_{5,1,-2}[\tilde{\mu},\,m_2]\Bigg)
\end{aligned}
$$